Nico Deshler

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Research Interests

I am deeply fascinated by interdisciplinary projects that span the intersection of physics and computer science with an emphasis on quantum optics, estimation theory, and computational imaging. Historically, connections between these disciplines have birthed rich spaces for scientific exploration and technological development.

Education

PhD, Optical Sciences 2021 - *pres.*
Wyant College of Optical Sciences
University of Arizona

BA, Physics and Computer Science 2016 - 2021
Department of Physics + Electrical Engineering and Computer Science
UC Berkeley

Research Experience

Intelligent Imaging and Sensing Lab | PI: Amit Ashok, University of Arizona 2021 - *pres.*

- Determined the quantum information limits of exoplanet *detection* and *localization* prescribed by the quantum Chernoff exponent and quantum Fisher information matrix (QFIM), respectively.
- Developed a Kraus operator formalism for modeling coronagraphs as lossy bosonic channels, enabling rigorous comparative performance surveys for state-of-the-art exoplanet imaging systems.
- Designed, built, and tested a quantum-optimal coronagraph using an SLM-based spatial mode sorter which accurately localized an exoplanet below the diffraction limit at $10^3 : 1$ star-planet contrast contrast. [patent link]
- Developed and deployed a hybrid SPADE + Direct Imaging instrument on the Lick observatory's Shane 3m telescope for super-resolution astronomical imaging.

Center for Quantum Networks | PI: Saikat Guha, University of Maryland 2021 - *pres.*

- Derived the QFIM for simultaneous estimation of mean photon numbers in ensembles of subdiffraction incoherent thermal emitters using CV Gaussian state formalism.
- Developed an adaptive SPADE-based measurement protocol for brightness estimation that outperforms direct imaging by diagonalizing a multimode classically-correlated thermal state with a Williamson decomposition.
- Developed a protocol for distributed magnetic field sensing with an ensemble of sub-diffraction nitrogen-vacancy color centers in diamond quantum that outperforms the conventional optically-detected magnetic resonance protocol.

Quantum Optomechanics Lab | PI: Dalziel Wilson, University of Arizona 2023 - 2026

- Introduced an interaction Hamiltonian for multimode cavity-free quantum optomechanics with spatially-structured probe states and conducted a Fisher information analysis for dynamical imaging of an optomechanical membrane.

Computational Imaging Lab | PI: Laura Waller, UC Berkeley 2017 - 2021

- Developed a prototype multi-sensor mask-based computational camera for synthetic large-format sensing. I implemented regularized inverse algorithms and novel calibration methods to address structured erasures inherent to measurements made with disjoint sensor arrays.
- Built ray-trace simulation pipelines in Python, Matlab, and Zemax OpticStudio for numerically assessing the reconstruction fidelity of a multi-sensor diffuser-based lensless camera.

Internships

Advanced Technology Center + Optical Center of Excellence | PI: David Theil, LM Space 2018 - 2020

- Implemented a signal processing chain for a PIC-based two-dimensional interferometric imaging system. I applied compressive sensing techniques with regularized loss functions to solve a sparsity-constrained inverse problem.
- Developed a multi-baseline captured using a PIC architecture with coupled imaging and laser communications capabilities. [patent link]
- Wrote user-defined surfaces in C for Zemax OpticStudio to facilitate structural and thermal analysis of optical components informed by finite-element simulations (surface deformation and index gradients).
- Developed an ultraviolet decontamination wand during the covid-19 pandemic with real-time surface dosage monitoring for targeted sterilization of frequently-touched surfaces. [patent link]

Whitesides Research Group | PI: George Whitesides, Harvard University 2017

- Developed a high-throughput cost-effective medical diagnostic instrument based on optical imaging of magnetically levitated samples for cell density characterization and quantification. [patent link]
- Prototyped optical assemblies using SolidWorks and 3D printing to realize instrument compatibility with point-of-care medical equipment (e.g. 96-well plates and flatbed scanners).

Flow Physics and Computation Lab | PI: Rajat Mittal, Johns Hopkins University 2015

- Studied the aeromechanics of long jumps in crickets as inspiration for novel forms of mechanical locomotion. []
- Employed photogrammetry techniques in Matlab to gather data on limb orientation throughout jump trajectories.
- Analyzed and modeled restoration torques produced by drag forces acting about the center of mass of the insect.

Publications

Journal Articles

2025 **N. Deshler**, I. Ozer, A. Ashok, S. Guha, "*Experimental demonstration of a quantum-optimal coronagraph using spatial mode sorters*," *Optica*, vol. 12, iss. 4, p. 518-529. [paper link]

2025 **N. Deshler**, S. Haffert, A. Ashok, "*Quantum limits of exoplanet detection and localization*," *Phys. Rev. A*, vol. 113, iss. 4, p. 042406. [paper link]

2025 C. Pluchar, W. He, J. Manley, **N. Deshler**, S. Guha, D. Wilson, "*Imaging-based quantum optomechanics*," *Phys. Ref. Lett.*, vol. 135, iss. 2, p. 023601. [paper link]

2025 A.K. Jha, S. Fleming, **N. Deshler**, A. Sajjad, M. Neifeld, A.Ashok, S. Guha, "*Multi-aperture telescopes at the quantum limit of superresolution imaging: Detecting subRayleigh object near a star*," *Acta. Astro.*, vol. 226, iss. 0094-5765, p. 531-541. [paper link]

2018 S. Ge, Y. Wang, **N. Deshler**, D.Preston, G. M. Whitesides, "*High-throughput density measurement using magnetic levitation*," J. Am. Chem. Soc., vol. 140, iss. 24, p. 7510–7518. [paper link]

Preprints

2025 **N. Deshler**, D. Daly, A. Majumder, K. Saha, S. Guha, "*Quantum limited spatial resolution of NV-diamond magnetometry*," arXiv:2508.13438 (Conditionally Accepted to Optica Quantum). [paper link]

Conference Proceedings & Presentations

Optica COSI 2026 Maastricht, Netherlands
Quantum limits of brightness estimation
N. Deshler, S. Ahire, S. Guha

SPIE Astronomical Telescopes + Instrumentation 2026 — Copenhagen, Denmark
Quantum limits of exoplanet discovery around resolved stars
N. Deshler, A. Ashok
Conference Abstract

Optica COSI 2024 — Toulouse, France
Experimental demonstration of a quantum-optimal direct imaging coronagraph
N. Deshler, I. Ozer, A. Ashok, S. Guha
Conference Abstract + Presentation Video

Optica COSI 2024 — Toulouse, France
Toward quantum resolution limit of magnetic field imaging with nitrogen-vacancy centers
N. Deshler, A. Majumder, K, Saha, S. Guha
Conference Abstract + Presentation Video

Optica COSI 2023 — Boston MA, USA
Quantum-based superresolution imaging with multi-aperture systems
N. Deshler, S. Guha, A. Ashok
Conference Abstract + Presentation Video

Optica COSI 2019 — Washington DC, USA
Multi-sensor lensless imaging: synthetic large-format sensing with a disjoint sensor array
E. Zhao, **N. Deshler**, K. Monakhova, L. Waller
Conference Abstract

APS Division of Fluid Dynamics 2017 — Tampa FL, USA
Aeromechanics of long jumps in spider crickets: Insights from experiments and modeling
E. Palmer, **N. Deshler**, R. Mittal
Conference Abstract

Awards & Fellowships

2023-2026	**NSF GRFP at University of Arizona** [award link]	*National Science Foundation*
2025	**Student Innovator of the Year** [award link]	*University of Arizona*
2018	**NSF CUR at UC Berkeley**	*National Science Foundation*
2018	**NSF REU at UC Berkeley** [award link]	*National Science Foundation*
2017	**NSF REU at Harvard University** [award link]	*National Science Foundation*
2015	**NSF RAHSS at Johns Hopkins University**	*National Science Foundation*

Skills

Programming	**Experiment**	**Human Languages**
Python, Matlab, C, HTML C++, Java, RISC-V, PyTorch, Slurm GLSL	Zemax, CAD/SolidWorks, LabView	English, Spanish

References

Amit Ashok (PhD Advisor)
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